

MEMORIA PRÁCTICA 1

SISTEMAS INFORMÁTICOS II

Pablo Gómez-Diego cuadrado

EJERCICIO 1:

Desplegamos la web localmente

```
((venv) ) [pbl@hermes] [/dev/pts/4] [main]
L[~/Documents/si2/si2_alumnos/P1-base]> python3.12 manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
February 16, 2026 - 10:42:15
Django version 5.2.2, using settings 'visaSite.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.

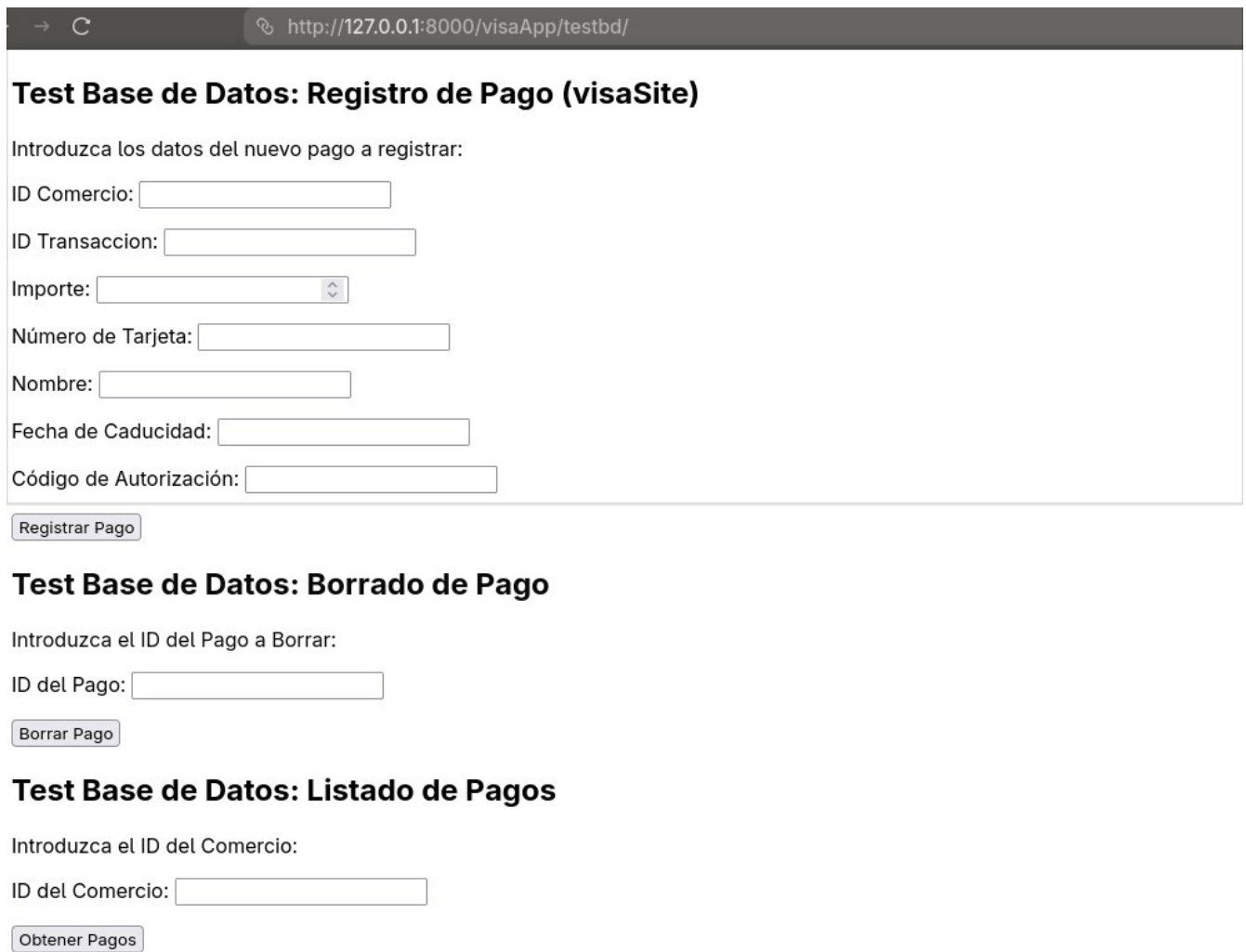
WARNING: This is a development server. Do not use it in a production setting. Use a produc
tion WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howt
o/deployment/
[16/Feb/2026 10:42:21] "GET /visaApp/ HTTP/1.1" 200 1115
[16/Feb/2026 10:42:29] "GET /visaApp/tarjeta/ HTTP/1.1" 200 1115
[16/Feb/2026 10:42:39] "GET /visaApp/pago/ HTTP/1.1" 200 940
[16/Feb/2026 10:42:48] "GET /visaApp/testbd/ HTTP/1.1" 200 2548
```

- Contenido del fichero env:

```
env x
8 # IMPORTANT: this file should not be in a repository
7 # To remove a file named env from a Git repository
6 # but keep it in the source (local system), follow these steps:
5 # Remove the file from Git tracking but keep it locally
4 ## git rm --cached env
3 # Add 'env' to .gitignore (so it's not tracked again)
2 ## echo "env" >> .gitignore
1 # Commit the changes
9 ## git commit -m "Removed env from Git tracking and added to .gitignore"
1 # Push the changes to the remote repository
2 ## git push
3 # use sqlite 3
4 ##DATABASE_SERVER_URL=sqlite:///db.sqlite3
5 # use postgres
6 DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@localhost:15432/si2db'
7 # The client does not need to store data in any database
8 # so let us define a sqlite in orden to avoid warning messages
9 DEBUG=True
10 SECRET_KEY = 'django-insecure-alczftn)j1#v%xmk@5j(n*px43c8kxgi_ua4%khc+t7g_)s9d'
```

- Respuesta del servidor al solicitar los endpoints:

-Endpoint testbd:



→ C http://127.0.0.1:8000/visaApp/testbd/

Test Base de Datos: Registro de Pago (visaSite)

Introduzca los datos del nuevo pago a registrar:

ID Comercio:

ID Transaccion:

Importe:

Número de Tarjeta:

Nombre:

Fecha de Caducidad:

Código de Autorización:

Test Base de Datos: Borrado de Pago

Introduzca el ID del Pago a Borrar:

ID del Pago:

Test Base de Datos: Listado de Pagos

Introduzca el ID del Comercio:

ID del Comercio:

-Endpoint pago:

Registro de Pago (visaSite)

Introduzca los datos del nuevo pago a registrar:

ID Comercio:

ID Transaccion:

Importe:

-Endpoint tarjeta:

Introduzca la Información de la Tarjeta de la Persona que Paga (visaSite)

Número de Tarjeta:

Nombre:

Fecha de Caducidad:

Código de Autorización:

EJERCICIO 2

- Resultado de los tests:

```
((venv) ) r[pbl@hermes] [/dev/pts/0] [main]
^[[~/Documents/si2/si2_alumnos/P1-base]> python3.12 manage.py test
Found 18 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
...Error: Registrando pago: null value in column "tarjeta_id" of relation "pago" violates not-null constraint
DETAIL: Failing row contains (5, C001, T001, 123, 2026-02-18 18:13:02.968789+00, 000, null).

.....
-----
Ran 18 tests in 0.090s

OK
Destroying test database for alias 'default'...
```

EJERCICIO 3

Desplegamos la aplicación usando las máquinas virtuales

- Estado del servicio de gunicorn en vm2:

```
si2@si2-ubuntu-vm-2:~/repo/plbase/P1-ws-backend$ sudo systemctl status gunicorn
● gunicorn.service - Gunicorn WSGI Application Server
   Loaded: loaded (/etc/systemd/system/gunicorn.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-02-22 22:37:11 CET; 1h 34min ago
     Main PID: 17064 (gunicorn)
        Tasks: 2 (limit: 1670)
      Memory: 56.0M (peak: 56.2M)
         CPU: 1.952s
    CGroup: /system.slice/gunicorn.service
            └─17064 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:ap
              └─17068 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:ap
```

- Contenido del fichero env:

```

18 # but keep it in the source (local system), follow these steps:
17 # Remove the file from Git tracking but keep it locally
16 ## git rm --cached env
15 # Add 'env' to .gitignore (so it's not tracked again)
14 ## echo "env" >> .gitignore
13 # Commit the changes
12 ## git commit -m "Removed env from Git tracking and added to .gitignore"
11 # Push the changes to the remote repository
10 ## git push
9 # use sqlite 3
8 ##DATABASE_SERVER_URL=sqlite:///db.sqlite3
7 # use postgres
6 DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@localhost:15432/si2db'
5 #DATABASE_SERVER_URL='postgresql://neondb_owner:npg_r1D00G9xXSAJ@ep-round-bonus-agdfmsb0-pooler.c-2.eu-central-1.aws.neon.tech/neondb?sslmode=require&channel_binding=require'
4 # The client does not need to store data in any database
3 # so let us define a sqlite in orden to avoid warning messages
2 DEBUG=True
1 SECRET_KEY = 'django-insecure-alczfthnjl1#v%xmkk05j(n*px43c8kxgi_ua4%khc+t7g_)s9d'

```

- Registro de pago exitoso:



<http://127.0.0.1:28000/visaApp/pago/>

Pago Registrado con Éxito (visaSite)

Id: 4

Codigo Respuesta: 000

Marca Tiempo : Feb. 18, 2026, 8:50 p.m.

Id Comercio: Mercado

Id Transaccion: 12345

Importe: 10.0

EJERCICIO 4

Ejecución del script de lectura

1. Con lectura en base de datos local:

```
((venv) ) r[pbl@apolo] [/dev/pts/3] [main ⚡]
^[~/Documents/SI2/sistemasInf2/P1-base]> ./metrics.sh read_1000_entries_from_db.py

| Iteration | Time elapsed (seg) |
|:-----:|:-----:|
| 1 | 0.653029 |
| 2 | 0.628659 |
| 3 | 0.635294 |
| 4 | 0.628110 |
| 5 | 0.619487 |
| 6 | 0.633327 |
| 7 | 0.619084 |
|:-----:|:-----:|

-----Stats resume:-----
| Mean | Standard Deviation |
|:-----:|:-----:|
| 0.630999 | 0.011529 |
|:-----:|:-----:|

((venv) ) r[pbl@apolo] [/dev/pts/3] [main ⚡]
^[~/Documents/SI2/sistemasInf2/P1-base]>

# env x
13 # but keep it in the source (local system), follow these steps:
12 # Remove the file from Git tracking but keep it locally
11 ## git rm --cached env
10 # Add 'env' to .gitignore (so it's not tracked again)
9 ## echo "env" >> .gitignore
8 # Commit the changes
7 ## git commit -m "Removed env from Git tracking and added to .gitignore"
6 # Push the changes to the remote repository
5 ## git push
4 # use sqlite 3
3 ##DATABASE_SERVER_URL=sqlite:///db.sqlite3
2 # use postgres
1 HOST='192.168.1.49'
14 DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@192.168.1.49:15432/si2db'
1 #DATABASE_SERVER_URL='postgresql://neondb_owner:npg_rid00G9xXSAJ@ep-round-bonus-agdfmsb0-pooler.c-2.eu-central-1.aws.neon.t
  ech/neondb?sslmode=require&channel_binding=require'
2 # The client does not need to store data in any database
3 # so let us define a sqlite in orden to avoid warning messages
4 DEBUG=True
5 SECRET_KEY = 'django-insecure-alczftnj1#$v%xmkk@5j(n*px43c8kxgi_ua4%khc+t7g_)s9d'
6
```

Al estar en una red local virtualizada, la latencia es mínima. El tiempo se consume principalmente en el procesamiento de la base de datos y el tráfico TCP/IP básico.

2. Lectura en base de datos de neon:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-base]> ./metrics.sh read_1000_entries_from_db.py

| Iteration | Time elapsed (seg) |
|:-----:|:-----:|
| 1 | 30.426202 |
| 2 | 31.023429 |
| 3 | 33.733130 |
| 4 | 33.641497 |
| 5 | 31.193128 |
| 6 | 28.592641 |
| 7 | 35.538664 |
|:-----:|:-----:|

-----Stats resume:-----
| Mean | Standard Deviation |
|:-----:|:-----:|
| 32.021242 | 2.377285 |
|:-----:|:-----:|

# env x
1 # but keep it in the source (local system), follow these steps:
2 # Remove the file from Git tracking but keep it locally
3 ## git rm --cached env
4 # Add 'env' to .gitignore (so it's not tracked again)
5 ## echo "env" >> .gitignore
6 # Commit the changes
7 ## git commit -m "Removed env from Git tracking and added to .gitignore"
8 # Push the changes to the remote repository
9 ## git push
10 # use sqlite 3
11 ##DATABASE_SERVER_URL=sqlite:///db.sqlite3
12 # use postgres
13 HOST='192.168.1.49'
14 #DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@192.168.1.49:15432/si2db'
15 DATABASE_SERVER_URL='postgresql://neondb_owner:npg_rid00G9xXSAJ@ep-round-bonus-agdfmsb0-pooler.c-2.eu-central-1.aws.neon.tech/neondb?sslmode=require&channel_binding=require'
16 # The client does not need to store data in any database
17 # so let us define a sqlite in order to avoid warning messages
18 DEBUG=True
19 SECRET_KEY = 'django-insecure-alczftnjl1$%v%mk@5j(n*px43c8kxgi_ua4%khc+t7g_)s9d'
20
```

El tiempo se dispara drásticamente a unos 32.02 s. Esto supone ser unas 50 veces más lento que el caso local. La razón técnica es el RTT, ya que al realizar 1000 consultas individuales una a una, estamos obligando al sistema a realizar 1000 viajes de ida y vuelta por internet. Incluso con una latencia baja, el acumulado penaliza enormemente el rendimiento.

3. Se accede mediante ORM:

Iteration	Time elapsed (seg)
1	0.884024
2	0.820781
3	0.863834
4	0.839534
5	0.835767
6	0.829454
7	0.839675

```
-----Stats resume:-----
Mean      | Standard Deviation |
:-----:  | :-----:          |
0.844724   | 0.021790           |
-----:-----
```

```
(venv) [pbl@apolo] [/dev/pts/3] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-base]>
```

```
read_1000_entries... x
13 import os
12 import time
11
10 import django  # Import "django" could not be resolved
9
8 os.environ.setdefault("DJANGO_SETTINGS_MODULE", "visaSite.settings")
7 django.setup()
6
5 from visaApp.models import Tarjeta
4
3 try:
2     lista_numeros = list(Tarjeta.objects.values_list("numero", flat=True)[:1000])
1
14 if not lista_numeros:
1     print("Error: No hay datos en la tabla tarjeta.")
2     start_time = time.time()
3     for num in lista_numeros:
4         Tarjeta.objects.get(pk=num)
5
6     end_time = time.time()
7
8     print(
9         f"Tiempo invertido en buscar las 1000 entradas una a una (ORM): {end_time - start_time:.6f} segundos"
10    )
11
12 except Exception as e:
13     print(f"Error: {e}")
```

El tiempo aumenta respecto al SQL puro a unos 0.84 s. Esto representa una sobrecarga de aproximadamente el 34%.

Esta diferencia confirma que el ORM introduce un coste de abstracción inevitable; por cada consulta, Django debe instanciar un objeto de la clase Tarjeta y mapear los tipos de datos de la base de datos a atributos de Python. Aunque el ORM mejora la productividad del desarrollo, penaliza el rendimiento en operaciones masivas de lectura individual debido al consumo de CPU necesario para la gestión de estos objetos.

Cuestión 1:

Considerando los datos obtenidos, 7 repeticiones son suficientes para obtener una estimación fiable en entornos locales, pero resultan escasas para el entorno en la nube. En el Caso 1, la desviación estándar es mínima ($\sigma \approx 0.011$ s), lo que indica una variabilidad insignificante y resultados muy consistentes. Sin embargo, en el Caso 2 (Neon.tech), la desviación aumenta considerablemente ($\sigma \approx 2.37$ s) debido al "jitter" o fluctuaciones del tráfico de Internet. Aunque 7 muestras permiten identificar la magnitud del retardo en la nube, se necesitaría un tamaño de muestra mayor para estabilizar la media en un entorno con tanta variabilidad de red.

Ejercicio 5

- Ejecución de los tests

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
L[~/Documents/SI2/sistemasInf2/P1-base]> python12 manage.py test
Found 18 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
...Error: Registrando pago: null value in column "tarjeta_id" of relation "pago" violates not-null constraint
DETAIL: Failing row contains (5, C001, T001, 123, 2026-02-19 14:39:57.810538+00, 000, null).

.....
-----
Ran 18 tests in 0.201s

OK
Destroying test database for alias 'default'...
((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
L[~/Documents/SI2/sistemasInf2/P1-base]> 
```

Ejercicio 6

- Ejecución de tests

```

((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]> python12 manage.py test visaAppWSBackend.test_views -v 2
Found 6 test(s).
Creating test database for alias 'default' ('test_si2db')...
Operations to perform:
  Synchronize unmigrated apps: messages, rest_framework, staticfiles
  Apply all migrations: admin, auth, contenttypes, sessions, visaAppWSBackend
Synchronizing apps without migrations:
  Creating tables...
  Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
  Applying visaAppWSBackend.0001_initial... OK
System check identified no issues (0 silenced).
test_01_tarjeta_check_valid_data (visaAppWSBackend.test_views.ApiViewTest.test_01_tarjeta_check_valid_data) ... ok
test_02_tarjeta_check_invalid_data (visaAppWSBackend.test_views.ApiViewTest.test_02_tarjeta_check_invalid_data) ... ok
test_05_list_pagos (visaAppWSBackend.test_views.ApiViewTest.test_05_list_pagos) ... ok
test_10_pago_store (visaAppWSBackend.test_views.ApiViewTest.test_10_pago_store) ... ok
test_20_delete_existing_pago (visaAppWSBackend.test_views.ApiViewTest.test_20_delete_existing_pago)
Test deleting an existing Pago object. ... ok
test_21_delete_nonexistent_pago (visaAppWSBackend.test_views.ApiViewTest.test_21_delete_nonexistent_pago)
Test attempting to delete a Pago object that does not exist. ... ok

-----
Ran 6 tests in 0.061s

OK
Destroying test database for alias 'default' ('test_si2db')...

```

Ejercicio 7

Conexión a los distintos endpoints. Nos conectamos mediante curl por conveniencia a la hora de adjuntar y recibir contenido.

-Caso de tarjeta inválida:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main]
L[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X POST http://192.168.1.49:28000/restapiserver/tarjeta/ \
-H "Content-Type: application/json" \
-d '{
  "numero": "0000 1111 2222 3333",
  "nombre": "Usuario Falso",
  "fechaCaducidad": "12/99",
  "codigoAutorizacion": "000"
}'
HTTP/1.1 404 Not Found
Server: gunicorn
Date: Sun, 22 Feb 2026 20:56:33 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, OPTIONS
X-Frame-Options: DENY
Content-Length: 54
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"message":"Datos no encontrados en la base de datos"}%
```

-Caso de tarjeta válida:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main]
L[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X POST http://192.168.1.49:28000/restapiserver/tarjeta/ \
-H "Content-Type: application/json" \
-d '{
  "numero": "3974 8228 3584 7557",
  "nombre": "Gabriel Vallejo Cozar",
  "fechaCaducidad": "07/29",
  "codigoAutorizacion": "861"
}'
HTTP/1.1 200 OK
Server: gunicorn
Date: Sun, 22 Feb 2026 20:56:46 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, OPTIONS
X-Frame-Options: DENY
Content-Length: 51
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"message":"Datos encontrados en la base de datos"}%
```

-Caso de pago inválido:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X POST http://192.168.1.49:28000/restapiserver/pago/ \
-H "Content-Type: application/json" \
-d '{
  "idComercio": "Tienda_Informatica",
  "idTransaccion": "TX-002",
  "importe": 50.00,
  "tarjeta_id": "0000 0000 0000 0000"
}'
HTTP/1.1 404 Not Found
Server: gunicorn
Date: Sun, 22 Feb 2026 20:58:50 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, DELETE, OPTIONS
X-Frame-Options: DENY
Content-Length: 40
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"message":"Error al registrar el pago"}%`
```

-Caso de registro de pago exitoso:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X POST http://192.168.1.49:28000/restapiserver/pago/ \
-H "Content-Type: application/json" \
-d '{
  "idComercio": "Mercado",
  "idTransaccion": "TX-001",
  "importe": 150.75,
  "tarjeta_id": "3974 8228 3584 7557",
  "codigoRespuesta": "000"
}'
HTTP/1.1 200 OK
Server: gunicorn
Date: Sun, 22 Feb 2026 21:00:06 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, DELETE, OPTIONS
X-Frame-Options: DENY
Content-Length: 178
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"id":6,"idComercio":"Mercado","idTransaccion":"TX-001","importe":150.75,"tarjeta":"3974 8228 3584 7557","marcaTiempo":"2026-02-22T22:00:06.706380+01:00","codigoRespuesta":"000"}%`
```

-Caso de comercio inválido:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X GET http://192.168.1.49:28000/restapiserver/comercio/Comercio_Fantasma_99
9/
HTTP/1.1 404 Not Found
Server: gunicorn
Date: Sun, 22 Feb 2026 21:01:43 GMT
Connection: close
Content-Type: text/html; charset=utf-8
X-Frame-Options: DENY
Content-Length: 3431
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin
```

-Caso de comercio válido:

```
((venv) ) r[pbl@apolo] [/dev/pts/3] [main]
L[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X GET http://192.168.1.49:28000/restapiserver/comercio/Mercado
HTTP/1.1 200 OK
Server: gunicorn
Date: Sun, 22 Feb 2026 21:03:29 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: GET, HEAD, OPTIONS
X-Frame-Options: DENY
Content-Length: 339
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

[{"id":4,"idComercio":"Mercado","idTransaccion":"12345","importe":10.0,"tarjeta":"23","marcaTiempo":"2026-02-18T20:50:41.837554+01:00","codigoRespuesta":"000"}, {"id":6,"idComercio":"Mercado","idTransaccion":"TX-001","importe":150.75,"tarjeta":"3974 8228 3584 7557","marcaTiempo":"2026-02-22T22:00:06.706380+01:00","codigoRespuesta":"000"}]
```

-Caso de borrado de pago incorrecto:

```
((venv) ) r[pbl@apolo] [/dev/pts/3] [main]
L[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X DELETE http://192.168.1.49:28000/restapiserver/pago/'100'
HTTP/1.1 404 Not Found
Server: gunicorn
Date: Sun, 22 Feb 2026 21:05:39 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, DELETE, OPTIONS
X-Frame-Options: DENY
Content-Length: 37
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"message":"Error al borrar el pago"}]
```

-Caso de borrado de pago exitoso:

```
((venv) ) r[pbl@apolo] [/dev/pts/3] [main]
L[~/Documents/SI2/sistemasInf2/P1-ws-backend]> curl -i -X DELETE http://192.168.1.49:28000/restapiserver/pago/'1'
HTTP/1.1 200 OK
Server: gunicorn
Date: Sun, 22 Feb 2026 21:05:06 GMT
Connection: close
Content-Type: application/json
Vary: Accept, Cookie
Allow: POST, DELETE, OPTIONS
X-Frame-Options: DENY
Content-Length: 40
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin

{"message":"Pago borrado correctamente"}]
```

Ejercicio 8:

- Resultado de los tests

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
^[[~/Documents/SI2/sistemasInf2/P1-ws-frontend]> python12 manage.py test visaAppWSFrontend.test_views -v 2
Found 8 test(s).
Creating test database for alias 'default' ('test_si2db')...
Operations to perform:
  Synchronize unmigrated apps: messages, staticfiles, visaAppWSFrontend
  Apply all migrations: admin, auth, contenttypes, sessions
Synchronizing apps without migrations:
  Creating tables...
  Running deferred SQL...
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
System check identified no issues (0 silenced).
test_015_aportarinfo_tarjeta_invalid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_015_aportarinfo_tarjeta_invalid_post)
Test invalid tarjeta submission. ... ok
test_01_aportarinfo_tarjeta_valid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_01_aportarinfo_tarjeta_valid_post)
Test valid tarjeta submission. ... ok
test_02_store_pago_valid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_02_store_pago_valid_post)
Test valid pago creation via API. ... ok
test_035_delpago_invalid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_035_delpago_invalid_post)
Test deletion of a non-existing pago. ... ok
test_03_delpago_valid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_03_delpago_valid_post)
Test deletion of an existing pago. ... ok
test_04_getPagos_post (visaAppWSFrontend.test_views.PagoViewsTest.test_04_getPagos_post)
Test retrieval of pagos filtered by comercio. ... ok
test_10_testdb_post (visaAppWSFrontend.test_views.PagoViewsTest.test_10_testdb_post)
Test combined tarjeta + pago submission. ... ok
test_11_testdb_invalid_post (visaAppWSFrontend.test_views.PagoViewsTest.test_11_testdb_invalid_post)
Test testdb endpoint with missing tarjeta data. ... ok

-----
Ran 8 tests in 0.410s

OK
Destroying test database for alias 'default' ('test_si2db')...
```

Ejercicio 9:

Desplegamos las aplicaciones web P1-ws-frontend y P1-ws-backend.

- Status de gunicorn de ambas máquinas vm2(backend) y vm3(frontend):

```
si2@si2-ubuntu-vm-1:~/repo/plbase/P1-ws-backend$ sudo systemctl status gunicorn
● gunicorn.service - Gunicorn WSGI Application Server
   Loaded: loaded (/etc/systemd/system/gunicorn.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-02-22 22:37:11 CET; 1h Main PID: 17064 (gunicorn)
     Main PID: 17064 (gunicorn)
       Tasks: 2 (Limit: 1670)
      Memory: 56.0M (peak: 56.2M)
         CPU: 1.804s
    CGroup: /system.slice/gunicorn.service
            └─17064 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:app
            └─17066 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:app

feb 22 23:40:35 si2-ubuntu-vm-2 gunicorn[17068]: raise RuntimeError(
feb 22 23:40:35 si2-ubuntu-vm-2 gunicorn[17068]: RuntimeError: You called this URL via POST, but the URL doesn't end in a s
feb 22 23:41:31 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/tarjeta/
feb 22 23:41:31 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/pago/999999998
feb 22 23:41:37 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/tarjeta/
feb 22 23:41:37 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/pago/999999998
feb 22 23:42:06 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/tarjeta/
feb 22 23:42:06 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/pago/999999998
feb 22 23:44:01 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/tarjeta/
feb 22 23:44:01 si2-ubuntu-vm-2 gunicorn[17068]: Not Found: /restapiserver/pago/999999998
[Name: si2@pts/3] [Info]
```

```
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]: [2026-02-22 23:30:54 +0100] [3855] [INFO] Using worker: sync
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3856]: [2026-02-22 23:30:54 +0100] [3856] [INFO] Booting worker with pid: 3856
(venv) si2@si2-ubuntu-vm-3:~$ sudo systemctl status gunicorn
● gunicorn.service - Gunicorn WSGI Application Server
   Loaded: loaded (/etc/systemd/system/gunicorn.service; disabled; preset: enabled)
   Active: active (running) since Sun 2026-02-22 23:30:54 CET; 15min ago
     Main PID: 3855 (gunicorn)
       Tasks: 2 (Limit: 1670)
      Memory: 69.0M (peak: 69.3M)
         CPU: 652ms
    CGroup: /system.slice/gunicorn.service
            └─3855 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:ap
            └─3856 /home/si2/venv/bin/python3 /home/si2/venv/bin/gunicorn --workers 1 --bind 0.0.0.0:8000 visaSite.wsgi:ap

Feb 22 23:30:54 si2-ubuntu-vm-3 systemd[1]: Started gunicorn.service - Gunicorn WSGI Application Server.
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]: /home/si2/venv/lib/python3.12/site-packages/gunicorn/util.py:25: UserWarning
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]:   import pkg_resources
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]: [2026-02-22 23:30:54 +0100] [3855] [INFO] Starting gunicorn 20.1.0
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]: [2026-02-22 23:30:54 +0100] [3855] [INFO] Listening at: http://0.0.0.0:8000
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3855]: [2026-02-22 23:30:54 +0100] [3855] [INFO] Using worker: sync
Feb 22 23:30:54 si2-ubuntu-vm-3 gunicorn[3856]: [2026-02-22 23:30:54 +0100] [3856] [INFO] Booting worker with pid: 3856
Feb 22 23:31:25 si2-ubuntu-vm-3 gunicorn[3856]: Not Found: /favicon.ico
[Name: si2@pts/3] [Info]
```

- Contenido de ambos ficheros env:

```
((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]> cat env
# but keep it in the source (local system), follow these steps:
# Remove the file from Git tracking but keep it locally
## git rm --cached env
# Add 'env' to .gitignore (so it's not tracked again)
## echo "env" >> .gitignore
# Commit the changes
## git commit -m "Removed env from Git tracking and added to .gitignore"
# Push the changes to the remote repository
## git push
# use sqlite 3
## DATABASE_SERVER_URL=sqlite:///db.sqlite3
# use postgres
HOST='192.168.1.49'
DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@192.168.1.49:15432/si2db'
#DATABASE_SERVER_URL='postgresql://neondb_owner:npg_rID0069xXSAJ@ep-round-bonus-agdfmsb0-pooler.c-2.eu-central-1.aws.neon.tech/neon
db?sslmode=require&channel_binding=require'
# The client does not need to store data in any database
# so let us define a sqlite in orden to avoid warning messages
DEBUG=True
SECRET_KEY = 'django-insecure-alczftn)j1#%v%xmK@5j(n*px43c8kxgi_ua4%khc+t7g_)s9d'

((venv) ) [pbl@apolo] [/dev/pts/3] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-ws-backend]>

[pbl@apolo] [/dev/pts/8] [main ⚡]
[~/Documents/SI2/sistemasInf2/P1-ws-frontend]> cat env
DATABASE_SERVER_URL='postgres://alumnodb:alumnodb@192.168.1.49:15432/si2db'

RESTAPIBASEURL=http://192.168.1.49:28000/restapiserver/

DEBUG=True
SECRET_KEY='django-insecure-alczftn)j1#%v%xmK@5j(n*px43c8kxgi_ua4%khc+t7g_)s9d'
```

- Interacción con los endpoints:
 - Verificación de tarjeta

```
→ http://192.168.1.49:38000/visaApp/tarjeta/
```

Introduzca la Informacion de la Tarjeta de la Persona que Paga (visaSite)

Número de Tarjeta:

Nombre:

Fecha de Caducidad:

Código de Autorización:

-Registro de pago



Registro de Pago (visaSite)

Introduzca los datos del nuevo pago a registrar:

ID Comercio:

ID Transaccion:

Importe:



Pago Registrado con Éxito (visaSite)

Id: 13

Codigo Respuesta: 000

Marca Tiempo : 2026-02-22T23:32:15.495798+01:00

Id Comercio: Mercado

Id Transaccion: id3

Importe: 10.0

-Visualización de base de datos

→ C <http://192.168.1.49:38000/visaApp/testbd/>

Test Base de Datos: Registro de Pago (visaSite)

Introduzca los datos del nuevo pago a registrar:

ID Comercio:

ID Transaccion:

Importe:

Número de Tarjeta:

Nombre:

Fecha de Caducidad:

Código de Autorización:

Registrar Pago

Test Base de Datos: Borrado de Pago

Introduzca el ID del Pago a Borrar:

ID del Pago:

Borrar Pago

Test Base de Datos: Listado de Pagos

Introduzca el ID del Comercio:

ID del Comercio:

Obtener Pagos